1:**What is OSI model?**

OSI stands for open systems interconnection . OSI model contains 7 layers

Application layer - application layer is user interface layer . it works on some protocols like HTTP, HTTPS, SMTP

Presentation layer - presentation layer handles compression, encryption and decryption of data.

session layer - session layer coordinates and terminates conversation between the applications. it maintain sessions between the client and server.

transport layer - It is known as heart of OSI model. it is responsible for transferring the data across the networks . it uses protocols like TCP and UDP.

here the TCP header is added and data transfers into segments.

Network layer - the primary function of network layer is to move data into another layer. this is were IP address are applied for routing purposes. here segments

are transfers into packets by adding IP headers.

Data-link layer - this layer handles moving data into "IN AND OUT" of physical link in a network . it assures that error free data is been transferred into frames

by adding Ethernet header with mac address.

physical layer - It handles row data and responsible for sending computer bits from one device to another device across the network.

2:What is DHCP explain the DORA Process with parameters?

DHCP stands for dynamic host configuration protocol. it is used for assigning IP address dynamically for client or for assigning IP address automatically.

it follows a process called DORA

DORA process is

Discover - In discover message client broadcast a message with source Ip 0:0:0:0 , destination Ip 255:255:255:255 into the network source mac as client mac and destination mac as FFFFFFFFFFFF.

Offer -DHCP server provide an offer with destination mac as client mac. source mac as server mac address . source Ip address as server ip address .destination Ip address as 255:255:255:255

Request - when client receives the offer message from the DHCP server .it broadcast a gracious ARP to check in order to any host in the network with the same Ip address . if there is no response from ARP then request message is broadcasted to the server showing the acceptance of Ip address offered .

Acknowledgement - The DHCP server acknowledges the request made by the client by providing Ip address requested along with gateway Ip, subnet mask, DNS server Ip, lease times and few other attributes.

3:WHAT IS DNS AND TYPES OF OF DNS QUERIES?

DNS stands for domain name system. DNS translate domain names to ip address and ip address to domain names.

3 types of DNS quries

Non-Recursive DNS Query:

In a non-recursive query, the DNS resolver asks the DNS server for information without seeking further assistance. The server must provide an answer from its cache or return a referral to another server. This method relies on the client to perform additional lookups if necessary.

Recursive DNS Query:

In a recursive query, the DNS resolver takes full responsibility for retrieving the requested information. It queries other DNS servers on behalf of the client until it finds the answer or fails. This process simplifies the user experience, as the client receives a final answer without managing multiple queries.

Iterative DNS Query:

An iterative query allows the DNS resolver to ask a server for information and, if the server cannot provide it, to return a referral to another server. The resolver then queries the referred server in turn, continuing until it finds the answer. This method minimizes the load on the DNS servers and spreads out the query process.

4:When do DNS use tcp port ?

When transferring large DNS zone files between DNS servers, TCP is used to ensure reliable data transmission.
If a DNS response exceeds 512 bytes the server may indicate that the response should be retried using TCP. This ensures that all necessary data is transmitted without truncation.

5:TCP Three-Way Handshake?
In TCP threeway handshake client initiate the connection by sending a SYN packet to the server with following parameters-sequence number , window size , length , Mss, window scaling factor and sack premiant ,then server sends a SYN-ACK message along with same parameters by adding acknowledged number. once client receives SYN-ACK message from the server then client acknowledges the SYN send by the server .

6:HTTP Request Methods & Response Codes?
Common HTTP methods include GET, POST, PUT, DELETE. Response codes are 1XX for information, 2XX for succuss, 3XX for redirection, 4XX for client error, 5XX for server error.

7:About SSL Handshake?
Client initiates SSL handshake by sending a "CLIENTHELLO" message to the server .this message includes TLS version. client supported cipher suits and random number along with SNI (server name index) . in reply server sends "serverhello" message to client containing server SSL certificate . chosen cipher suits and random number . client verifies the server SSL certificates with the certificate authority. once certificate validation is done. client shares a premaster secret which is encrypted by public key and it can only decrypted by private key of the server. then both client and server generates session keys . after that client sends a finished message that encrypted by session keys . A secure symmetric encryption is achieved. then communication happens in secured channel.

8:WHAT ARE THE MODES IN IPSEC AND CONFIGERATIONS?
IPSEC has two modes: Transport mode (encrypts payload) and Tunnel mode (encrypts entire packet).
Tunnel mode - Protects data in network-to-network or site-to-site scenarios. It encapsulates and protects the entire IP packet-the payload including the original IP header and a new IP header (protects the entire IP payload including user data).
Transport mode - Protects data in host-to-host or end-to-end scenarios. In transport mode, IPsec protects the payload of the original IP datagram by excluding the IP header (only protects the upper-layer protocols of IP payload (user data)).
IPsec protocols AH and ESP can operate in either transport mode or tunnel mode.
In IPSEC configuration we have two phases. phases 1 and phase 2 .phase 1 is used for connection establishment. phase 2 is used for data transferring . in phase 1 we configure in two modes , one is main mode and another one is aggressive mode. phase 2 is for quick mode . In main mode we have six messages . in aggressive mode we have three messages . main mode is more secure then aggressive mode .
IPsec use port 500 .

9:Scenario 1-Troubleshooting Half-Loaded Page?
(when did you notice the issue ? , is it happening for all the URL or single URL ? ,Have you tried it in a different browser ?)
Check network connectivity with ping, verify DNS resolution with nslookup, examine browser console for errors, and check firewall settings that might block resources.

10:Scenario 2- Frequent network disconnection and slow internet speeds how will you troubleshoot?

11:What is Functionality of ping and trace route?

Ping: Sends ICMP Echo Request packets to a target IP address or hostname to test connectivity and measure round-trip time. It checks if the target is reachable and how long it takes for packets to travel to the destination and back.
Traceroute: Traces the path packets take to reach a target by sending packets with gradually increasing Time to Live (TTL) values. It identifies each hop along the route, showing the IP addresses and response times for each hop.
Scenario 3- IP-sec tunnel flapping frequently on a new configuration?
Check Configuration Settings,
network for stability issues. High latency, packet loss, or intermittent connectivity can cause the tunnel to flap. Use tools like ping or traceroute to test connectivity between the endpoints.
If the MTU is too high, packets may be fragmented, causing connectivity issues. Consider adjusting the MTU to a lower value on both ends.
Verify the keep alive and DPD settings to ensure they are properly configured.
Check firewall rules and NAT configurations on both sides to ensure that IP-sec traffic is allowed.

12:What Do you know about SAML and SAML Workflow?

SAML stands for security assertion makeup language . it is a protocol that allows users to access multiple applications with single set of credentials.
SAML Workflow:
In a typical SAML workflow, a user attempts to access a service provider, is redirected to the identity provider for authentication, and upon successful login, receives a SAML assertion that allows access to the requested service.